

RRR

Lab10 (Parteneriat cu laborile scrise de PETrascu)

Reglare $\mathcal{H}_2$ (o sa fac si o recalipulare la calcularea normei $\mathcal{H}_2$ si $\mathcal{L}_2$)

Pana sa intram in reglare $\mathcal{H}_2$, mai intai o sa vreau sa vorbesc despre norma $\mathcal{L}_2$ (spatiu Lebesgue) si $\mathcal{H}_2$ (spatiu Hardy)

Pentru **Matricea de Transfer (cazul MIMO)** definim $\|G\|_2 := \left\{ \frac{1}{2\pi} \int_{-\infty}^{\infty} \text{tr}[G^{\sim}(j\omega)G(j\omega)] d\omega \right\}^{\frac{1}{2}}$ care se numeste norma $\mathcal{L}_2$

$$\|G\|_2 < \infty \Leftrightarrow G \in \mathcal{RL}_2(s)^{p \times m} \Leftrightarrow \begin{cases} G(s) \text{ nu are poli pe } \mathbb{C}_0 \\ G(\infty) = 0, \text{ adica este strict proprie} \end{cases}$$

S-ar putea rezolva norma 2 folosind **Teoria Reziduurilor**, cel mai bine folosim unele formule pe stare (Lyapunov)

Presupunem ca sistemul este luat aleator si este reprezentat intr-o stare minimala. Norma

lui $\mathcal{L}_2$ este finita $\Leftrightarrow \begin{cases} D = 0 \\ A \text{ este dihotomica} \end{cases}$. In acest sens, sistemul se poate aduce la forma

$$G(s) = G_s(s) + G_a(s), \text{ unde } \begin{cases} G_s(s) \text{ are polii in } \mathbb{C}_- \text{ (este stabil) sau } G_s(s) \in \mathcal{RH}_2 \\ G_a(s) \text{ are polii in } \mathbb{C}_+ \text{ (este antistabil) sau } G_a(s) \in \mathcal{RH}_2^{\perp}. \end{cases} \text{ Atunci are loc rezultatul:}$$

$$\|G\|_2^2 = \|G_s\|_2^2 + \|G_a\|_2^2.$$

O mica observatie: Stim ca G_s si G_a sunt ortogonale si ca $\|\tilde{G}\|_2 = \|G\|_2$. In acest sens, daca G_a este antistabil, asta inseamna ca $\tilde{G}_a(s) = G_a^T(-s)$ care este stabil. Asadar, putem rescrie egalitatea in felul urmator:

$$\|G\|_2^2 = \|G_s\|_2^2 + \|\tilde{G}_a\|_2^2 \text{ si este suma de 2 stabile}$$

Acum trebuie calculate cele 2 norme de tip $\mathcal{H}_2$. Cum facem asta? Uite aici:

Pentru A stabil, norma 2 se calculeaza:

$$\|G\|_2 = [\text{tr}(B^T Q B)]^{\frac{1}{2}} = [\text{tr}(C P C^T)]^{\frac{1}{2}}, \text{ unde } P \text{ si } Q \text{ sunt solutiile unice ale ecuatiilor Lyapunov}$$

$$\begin{cases} AP + PA^T + BB^T = 0 \\ A^T Q + QA + C^T C = 0 \end{cases}$$

Pentru **A instabil, norma 2 se calculeaza:**

$$\|G\|_2 = [\text{tr}(B^T \hat{Q} B)]^{\frac{1}{2}} = [\text{tr}(C \hat{P} C^T)]^{\frac{1}{2}}, \text{ unde } \hat{P} \text{ si } \hat{Q} \text{ sunt solutiile unice ale ecuatiilor Lyapunov}$$

$$\begin{cases} A\hat{P} + \hat{P}A^T - BB^T = 0 \\ A^T \hat{Q} + \hat{Q}A - C^T C = 0 \end{cases}, \text{ unde } \text{tr}(A) \text{ reprezinta suma elementelor de pe diagonala principala}$$

Daca $\hat{P} \geq 0$, atunci perechea (A, B) este controlabila

Daca $\hat{Q} \geq 0$, atunci perechea (C, A) este observabila

Acum intram in reglare $\mathcal{H}_2$

$$\text{Fie sistemul generalizat } \begin{cases} \dot{x} = Ax + B_1 u_1 + B_2 u_2 \\ y_1 = C_1 x + D_{11} u_1 + D_{12} u_2 \\ y_2 = C_2 x + D_{21} u_1 + D_{22} u_2 \end{cases}$$

$$\text{Reprezentarea pe stare arata asa: } T = \begin{bmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{bmatrix} = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix}$$

$$\mathbf{T} = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & O & D_{12} \\ C_2 & D_{21} & O \end{array} \right] = \left[\begin{array}{cc} \mathbf{T}_{11} & \mathbf{T}_{12} \\ \mathbf{T}_{21} & \mathbf{T}_{22} \end{array} \right],$$

Problema optimala $\mathcal{H}_2$ arata asa: $\min_{K \text{ stabil}} \|TLFI(T, K)\|_2$, unde $TLFI(T, K) = T_{cl}$ (In slide-urile de curs este notat

$T_{y_1 u_1}$)

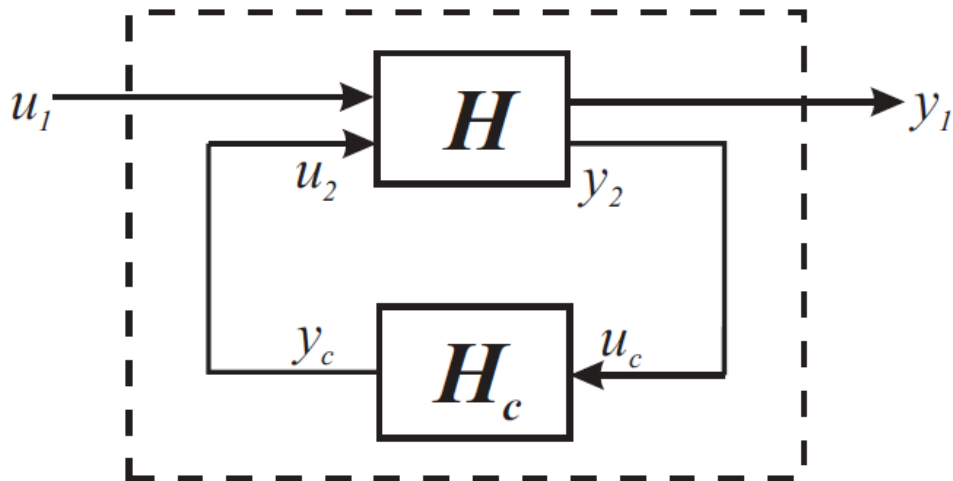


Figura 5: Transformare linear fracționară inferioară

$$TLFi(T, K) = T_{11} + T_{12}K(I - T_{22}K)^{-1}T_{21}$$

Fie $K = \begin{bmatrix} A_K & B_K \\ C_K & D_K \end{bmatrix}$. Trebuie sa existe $(I - T_{22}K)^{-1}$. Pentru a asigura buna definire, trebuie ca $\begin{bmatrix} I & -D_K \\ -D_{22} & I \end{bmatrix}$ sa fie inversabil

Sa zicem reprezentarea pe stare $TLFi(T, K) = T_{cl} = \begin{bmatrix} A_{cl} & B_{cl} \\ C_{cl} & D_{cl} \end{bmatrix}$

Ideea de baza este sa gasim un regulator K astfel incat $\Lambda(A_{cl}) \in \mathbb{C}_-$ (daca este satisfacuta asta, este

stabilizator) si sa aiba $\|T_{cl}\|_2^2 < \infty$, adica $\begin{cases} T_{cl}(\infty) = 0 \\ T_{cl}(s) \text{ sa NU aiba poli in } \mathbb{C}_0 \end{cases}$. Sa detaliez la $T_{cl}(\infty) = 0$, practic $D_{cl} = 0$.

Asta inseamna ca $D_{11} + D_{12}D_K(I - D_{22}D_K)^{-1}D_{21} = 0$

Ce presupuneri facem:

- (A, B_2) stabilizabil
- (C_2, A) detectabil
- $D_{11} = 0$ pentru a asigura ca avem mereu solutie
- $D_K = 0$ pentru a avea buna definire asigurata a TLFi-ului si pentru a avea solutie EMAR
- $D_{22} = 0$. De asta nu este neaparat nevoie, dar se poate presupune pentru a usura calculele. Asta zice ca TLFi-u este strict propriu

Prima si a treia presupunere ne zic ca norma 2 este finita si ca $T_{cl}(\infty) = D_{cl} = 0$. Din pacate, aceste



cerinte sunt DOAR pentru a asigura solutia problemei $\mathcal{H}_2$

Care-i faza cu $D_{22} = 0$:

Sa zicem ca avem $\tilde{T} = T - \begin{bmatrix} 0 & 0 \\ 0 & D_{22} \end{bmatrix}$

DESENE

De aici scoatem ca $u_2 = \tilde{K}(y_2 - D_{22}u_2)$. Vedem ca $u_2 = (I + \tilde{K}D_{22})^{-1} \tilde{K}y_2$, cu $\tilde{K}D_{22}$ este strict propriu.

$(I + \tilde{K}D_{22})^{-1} \tilde{K} = K$. Ideea e ca ne reducem mult calculele aici

O sa reatasez T pentru a ne ajuta mai departe: $T = \begin{bmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{bmatrix} = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix}$

$$\mathbf{T} = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & O & D_{12} \\ C_2 & D_{21} & O \end{array} \right] = \left[\begin{array}{cc} \mathbf{T}_{11} & \mathbf{T}_{12} \\ \mathbf{T}_{21} & \mathbf{T}_{22} \end{array} \right],$$

Fie sistemul nostru T cu 2 triplete Popov asociate: $\left\{ \begin{array}{l} \Sigma_{12} = \left(A; B_2; \begin{bmatrix} C_1^T C_1 & C_1^T D_{12} \\ D_{12}^T C_1 & D_{12}^T D_{12} \end{bmatrix} \right) \\ \Sigma_{21} = \left(A^T; C_2^T; \begin{bmatrix} B_1 B_1^T & B_1 D_{21}^T \\ D_{21} B_1^T & D_{21} D_{21}^T \end{bmatrix} \right) \end{array} \right\}$. Practic astea sunt

factorizarile inner-outer pentru T_{12} si co-inner-outer pentru T_{21} . La Σ_{12} are sens forma. La Σ_{21} pare mai

ciudatica. De ce? Noi stim 2 chestii: $\begin{cases} (C_2, A) \text{ detectabil} \Rightarrow (A^T, C_2^T) \text{ stabilizabil} \\ \Sigma = \left(A; B; \begin{bmatrix} Q & L \\ L^T & R \end{bmatrix} \right), \text{ unde } Q = C^T C, R = D^T D \text{ si } L = C^T D \end{cases}$ Practic la tripletul Popov Σ_{21} , $A = A^T$ si $B = C_2^T$. Asta inseamna ca $C = B_1^T$ si $D = D_{21}^T$ si se inlocuiesc in matrice.

Pornind de la cele 2 triplete Popov, facem urmatoarele presupuneri:

H1) Stim deja ca (A, B_2) stabilizabil. Totodata, mai presupunem ca $R_{12} = D_{12}^T D_{12} > 0$ are rang integ pe coloane si ca matricea $\begin{bmatrix} A - j\omega I & B_2 \\ C_1 & D_{12} \end{bmatrix}$ are rang integ pe coloane, adica $\text{rang} \begin{bmatrix} A - j\omega I & B_2 \\ C_1 & D_{12} \end{bmatrix} = n + m_2, \forall \omega \in \mathbb{R}$

H2) Stim deja ca (C_2, A) detectabil Totodata, mai presupunem ca $R_{21} = D_{21} D_{21}^T > 0$ are rang integ pe linii si ca matricea $\begin{bmatrix} A - j\omega I & B_1 \\ C_2 & D_{21} \end{bmatrix}$ are rang integ pe linii, adica $\text{rang} \begin{bmatrix} A - j\omega I & B_1 \\ C_2 & D_{21} \end{bmatrix} = n + p_2, \forall \omega \in \mathbb{R}$

Daca **H1)** si **H2)** au loc, atunci

- $\text{EMAR}(\Sigma_{12}) = A^T X + XA + Q - (XB + L)R^{-1}(B^T X + L^T)$ are solutie stabilizatoare $X_s \geq 0$:

$$A^T X_s + X_s A + C^T C - (X_s B + C_1^T D_{12})(D_{12}^T D_{12})^{-1}(B_2^T X_s + D_{12}^T C) = 0 \text{ si reactie stabilizatoare}$$

$$F_s := -R^{-1}(B^T X + L^T) = -(D_{12}^T D_{12})^{-1}(B_2^T X_s + D_{12}^T C)$$

- $\text{EMAR}(\Sigma_{21}) = A^T X + XA + Q - (XB + L)R^{-1}(B^T X + L^T)$ are solutie stabilizatoare $Y_s \geq 0$:

$$A^T Y_s + Y_s A + B_1 B_1^T - (Y_s C_2^T + B_1 D_{21}^T)(D_{21} D_{21}^T)^{-1}(C_2 Y_s + D_{21} B_1^T) = 0 \text{ si reactie stabilizatoare}$$

$$K_s := -R^{-1}(B^T X + L^T) = -(D_{21} D_{21}^T)^{-1}(C_2 Y_s + D_{21} B_1^T)$$

- Regulatorul optimal si strict propriu in sens $\mathcal{H}_2$ este dat de $K_{\text{opt}} = \begin{bmatrix} A + B_2 F_s + K_s^T C_2 & -K_s^T \\ F_s & 0 \end{bmatrix}$

$$K(s) = \begin{bmatrix} A + BF + LC + LDF & -L \\ F & 0 \end{bmatrix}, \text{ unde } \begin{cases} \Lambda(A + BF) \subset \mathbb{C}^- \\ \Lambda(A + LC) \subset \mathbb{C}^- \end{cases}. D_{22} \text{ si } D_{11} \text{ si } D_K \text{ am zis ca sunt 0. } \underline{\text{Chestia asta}}$$

merge DOAR DACA $D_{22} = 0$. Daca am $D_{22} \neq 0$, atunci fac bucla de reactie dintre $K_{\text{opt}}(s)$ si D_{22}

Valoarea optimala (minimala) a normei $\mathcal{H}_2$ este:

$$\min_{K \text{ stabil}} \|TLFi(T, K)\|_2 = \min_{K \text{ stabil}} \|T_{cl}\|_2 = [tr(B_1^T X_s B_1) + tr(W_x Y_s W_x^T)]^{\frac{1}{2}} = [tr(W_y X_s W_y^T) + (C_1 Y_s C_1^T)]^{\frac{1}{2}}. \text{ Ok, dar cine sunt}$$

$\begin{cases} W_x \\ W_y \end{cases}$? Ele ies din Sistemul Kalman-Yakubovich-Popov(SKYP). Aici o sa presupunem ca $J = I$. Avem 2 SKYP-uri, unul pentru Σ_{12} ai altul pentru Σ_{21} .

$$\begin{aligned} R &= V^T J V, \\ L + X B &= W^T J V, \\ Q + A^T X + X A &= W^T J W, \end{aligned}$$

Am zis de la inceput ca luam $J = I$. Deci putem rescrie SKYP asa: $\begin{cases} R = V^T V \\ L + X B = W^T V \\ A^T X + X A + Q = W^T W \end{cases}$

Acum sa luam SKYP(Σ_{12}, I). Avem $\begin{cases} D_{12}^T D_{12} = V_x^T V_x \\ C_1^T D_{12} + X_s B_2 = W_x^T V_x \\ C_1^T C_1 + A^T X_s + X_s A = W_x^T W_x \end{cases}$. Scoatem pe V_x ca fiind factor cholesky

din $D_{12}^T D_{12}$. De asemenea, noi stim cat este F_s . Totodata, stim asa: $F = -V^{-1}W$, unde F este rectia stabilizatoare din EMAR(Σ). Asadar, putem scrie ca $F_s = -V_x^{-1}W_s$. Noi stim pe V_x si F_s . Scoatem pe $W_s = -V_x F_s$

Acum sa luam SKYP(Σ_{21}, I). Avem $\begin{cases} D_{21} D_{21}^T = V_y^T V_y \\ B_1 D_{21}^T + Y_s C_2^T = W_y^T V_y \\ B_1 B_1^T + A^T Y_s + Y_s A = W_y^T W_y \end{cases}$. Scoatem pe V_y ca fiind factor cholesky

din $D_{21} D_{21}^T$. De asemenea, noi stim cat este K_s . Totodata, stim asa: $F = -V^{-1}W$, unde F este rectia stabilizatoare din EMAR(Σ). Asadar, putem scrie ca $K_s = -V_y^{-1}W_y$. Noi stim pe V_y si K_s . Scoatem pe $W_y = -V_y K_s$

Laborator

Remarca: matricile sunt alese aleator, deci la executarea scriptului o sa dea rezultate diferite

```
clear
close all
```

```
clc
```

```
A = randn(5); % sistem cu 5 stari  
B2 = randn(5,2); % m2 = 2  
C2 = randn(2,5); % p2 = 2  
D22 = randn(2,2); % Am zis ca nu e neaparat sa il iau 0 mereu. Vrem 0 pentru  
usurinta de calcul
```

```
B1 = randn(5,3); % m1 = 3  
C1 = randn(3,5); % p1 = 3
```

```
D11 = zeros(3,3);  
D12 = randn(3,2);  
D21 = randn(2,3);
```

Acum ca ne-am definit sistemul, sa ne apucam sa rezolvam problema $\mathcal{H}_2$. Mai intai, trebuie sa vedem ca matricele D_{12} are rang integ pe coloane, respectiv D_{21} are rang integ pe linii. Problema: nu am mereu garantie la matrici aleatoare ca o sa imi iasa cu rang integ pe linii/coloane

D21

```
D21 = 2x3  
-0.4189    0.8998    1.0294  
-0.1403   -0.3001   -0.3451
```

```
rank(D21) % se poate vedea ca matricea are rang integ pe linii
```

```
ans =  
2
```

D12

```
D12 = 3x2  
-0.6086   -1.3429  
-1.2226   -1.0322  
0.3165    1.3312
```

```
rank(D12) % se poate vedea ca am rang integ pe coloane
```

```
ans =  
2
```

Acum verific daca (A, B_2) este stabilizabil si ca (C_2, A) este detectabil. Cel mai ok ne uitam la zerourile de transmisie (bine, o sa vedem ca sistemul (A, B_2, C_2, D_{22}) este minimal, deci zerouri de transmisie = zerouri de invarianta)

```
tzero(A,B2,[],[]) % controlabil, deci si stabilizabil
```

```
ans =
```

```
0x1 empty double column vector
```

```
tzero(A,[],C2,[]) % observabil, deci si detectabil
```

ans =

0×1 empty double column vector

Acum trebuie sa vedem ca rangul matricilor $\text{rang} \begin{bmatrix} A - j\omega I & B_2 \\ C_1 & D_{12} \end{bmatrix} = n + m_2, \forall \omega \in \mathbb{R}$ si

$\text{rang} \begin{bmatrix} A - j\omega I & B_1 \\ C_2 & D_{21} \end{bmatrix} = n + p_2, \forall \omega \in \mathbb{R}$. Daca sunt goale, atunci nu inseamna neaparat ca au rang intreg pe coloane

```
tzero(A,B2,C1,D12) % Cu asta ne asiguram ca nu pica rangul matricii asteia
```

ans =

0×1 empty double column vector

```
tzero(A,B1,C2,D21) % La fel, sa nu pice rangul undeva
```

ans =

0×1 empty double column vector

Acum verificam rangul matricilor si ne asiguram ca $\text{rang} \begin{bmatrix} A - j\omega I & B_2 \\ C_1 & D_{12} \end{bmatrix} = n + m_2, \forall \omega \in \mathbb{R}$ si

$\text{rang} \begin{bmatrix} A - j\omega I & B_1 \\ C_2 & D_{21} \end{bmatrix} = n + p_2, \forall \omega \in \mathbb{R}$

```
rank([A,B2;C1,D12]) % 5 stari si m2 = 2
```

ans =

7

```
rank([A,B1;C2,D21]) % 5 stari si p2 = 2
```

ans =

7

Cu astea tin ipotezele. Adica putem rezolva cu EMAR

Mai intati ne uitam la tripletul Popov $\Sigma_{12} = \left(A; B_2; \begin{bmatrix} C_1^T C_1 & C_1^T D_{12} \\ D_{12}^T C_1 & D_{12}^T D_{12} \end{bmatrix} \right)$

```
Q12 = C1' * C1
```

Q12 = 5×5

5.2224	-4.0024	-1.2169	0.9295	0.6999
-4.0024	7.8843	2.2321	-3.1544	-0.8119
-1.2169	2.2321	3.3753	-4.1109	-0.4140
0.9295	-3.1544	-4.1109	5.2224	0.4726
0.6999	-0.8119	-0.4140	0.4726	0.1209

```
L12 = C1' * D12
```



```
L12 = 5x2
-2.2442 -1.2161
 3.9138  5.3278
 1.0467  2.4433
-1.4314 -3.5947
-0.4219 -0.4771
```

```
R12 = D12' * D12
```

```
R12 = 2x2
 1.9653  2.5005
 2.5005  4.6408
```

Acum vedem tripletul Popov $\Sigma_{21} = \left(A^T; C_2^T; \begin{bmatrix} B_1 B_1^T & B_1 D_{21}^T \\ D_{21} B_1^T & D_{21} D_{21}^T \end{bmatrix} \right)$

```
Q21 = B1 * B1'
```

```
Q21 = 5x5
 7.9910 -3.1254 -5.0530 -3.3334 -3.1976
-3.1254  1.4740  2.2330  1.3675  1.2395
-5.0530  2.2330  3.4989  2.0466  1.9538
-3.3334  1.3675  2.0466  1.7878  1.5026
-3.1976  1.2395  1.9538  1.5026  1.3573
```

```
L21 = B1 * D21'
```

```
L21 = 5x2
 1.9848 -0.7127
-1.3450  0.3675
-1.9408  0.6161
-0.6535  0.0938
-0.6257  0.1795
```

```
R21 = D21 * D21'
```

```
R21 = 2x2
 2.0448 -0.5665
-0.5665  0.2288
```

Acum aflam de la $\Sigma_{12} = \left(A; B_2; \begin{bmatrix} C_1^T C_1 & C_1^T D_{12} \\ D_{12}^T C_1 & D_{12}^T D_{12} \end{bmatrix} \right) X_s$ si F_s

```
[Xs,Fs,spectrus12] = icare(A,B2,Q12,R12,L12)
```

```
Xs = 5x5
 16.7987 -12.4916  1.5169 -14.4783 -24.0990
-12.4916  11.5903 -3.6821  7.0676  12.1485
 1.5169 -3.6821  3.4720  2.8363  0.9745
-14.4783  7.0676  2.8363  28.3900  29.2237
-24.0990  12.1485  0.9745  29.2237  70.3221
Fs = 2x5
 6.6231 -7.7418  3.4882 -10.7247 -7.9218
-4.3428  6.3636 -1.9868  5.1238  2.2252
spectrus12 = 5x1 complex
-0.4545 + 1.4504i
-0.4545 - 1.4504i
-0.6478 + 0.0000i
```

```
-2.3205 + 0.0000i
-4.5839 + 0.0000i
```

```
Fs = -Fs; % chestia asta pentru reactie pozitiva
eig(A + B2*Fs)
```

```
ans = 5x1 complex
-4.5839 + 0.0000i
-2.3205 + 0.0000i
-0.6478 + 0.0000i
-0.4545 + 1.4504i
-0.4545 - 1.4504i
```

```
eig(Xs) % verificam ca solutia Riccati este pozitiv semi-definita
```

```
ans = 5x1
0.0004
1.8203
13.6980
16.2006
98.8536
```

Acum aflam de la $\Sigma_{21} = \left(A^T; C_2^T; \begin{bmatrix} B_1 B_1^T & B_1 D_{21}^T \\ D_{21} B_1^T & D_{21} D_{21}^T \end{bmatrix} \right)$ pe Y_s si K_s

```
[Ys,Ks,spectrus21] = icare(A',C2',Q21,R21,L21)
```

```
Ys = 5x5
2.3484 -0.3441 -1.2594 -1.0471 -0.0069
-0.3441 8.9923 -2.6121 9.3430 -0.4145
-1.2594 -2.6121 1.7586 -2.0742 0.2039
-1.0471 9.3430 -2.0742 10.3122 -0.4527
-0.0069 -0.4145 0.2039 -0.4527 0.1933
Ks = 2x5
3.1945 2.6505 -3.5493 2.5960 -1.7562
11.2256 -4.4250 -6.9861 -8.7557 -4.2661
spectrus21 = 5x1 complex
-0.5577 + 1.5618i
-0.5577 - 1.5618i
-1.3498 + 0.0000i
-2.2865 + 0.0000i
-11.4313 + 0.0000i
```

```
Ks = -Ks'; % negarea am zis de ce. Transpunerea este de la dualitate (recomand
revedere curs TSA)
eig(A + Ks*C2) % na
```

```
ans = 5x1 complex
-11.4313 + 0.0000i
-0.5577 + 1.5618i
-0.5577 - 1.5618i
-2.2865 + 0.0000i
-1.3498 + 0.0000i
```

```
eig(Ys)
```

```
ans = 5x1
0.0083
0.1835
```

```

0.3236
3.4077
19.6817

```

Partea 1 din teorema merge

```
K_opt = ss(A + B2*Fs + Ks*C2, -Ks, Fs, 0) % cand D22 = 0
```

```
K_opt =
```

```

A =
      x1      x2      x3      x4      x5
x1  1.878  -10.98  0.7181  -7.851  15.59
x2   9.83   -7.511  1.481  -18.82  -12.33
x3  0.7384   1.72   1.044   4.427  -12.59
x4   6.09   1.351  0.3782  -13.98  -17.56
x5   2.368  -0.4948  1.837   1.048  -8.514

```

```

B =
      u1      u2
x1  3.194  11.23
x2  2.651  -4.425
x3 -3.549  -6.986
x4  2.596  -8.756
x5 -1.756  -4.266

```

```

C =
      x1      x2      x3      x4      x5
y1 -6.623  7.742  -3.488  10.72  7.922
y2  4.343  -6.364  1.987  -5.124  -2.225

```

```

D =
      u1  u2
y1   0   0
y2   0   0

```

Continuous-time state-space model.
Model Properties

```
K_opt_h2 = feedback(K_opt,D22) % cand D22 != 0
```

```
K_opt_h2 =
```

```

A =
      x1      x2      x3      x4      x5
x1  42.91  -46.3  25.28  -93.06  -62.69
x2  18.59  -23.14  4.835  -25.01  -10.37
x3 -34.76   35.5  -19.45  73.34  47.66
x4  6.527  -9.553  -1.817  0.7209  5.891
x5 -16.69  17.02  -9.307  38.95  25.24

```

```

B =
      u1      u2
x1  3.194  11.23
x2  2.651  -4.425
x3 -3.549  -6.986
x4  2.596  -8.756
x5 -1.756  -4.266

```

```

C =
      x1      x2      x3      x4      x5
y1 -6.623  7.742  -3.488  10.72  7.922
y2  4.343  -6.364  1.987  -5.124  -2.225

```

```
D =
      u1  u2
y1    0    0
y2    0    0
```

Continuous-time state-space model.
Model Properties

```
T = ss(A,[B1,B2],[C1;C2],[D11,D12;D21,D22]); % T nostru original
T_cl = lft(T,K_opt_h2)
```

```
T_cl =

A =
      x1      x2      x3      x4      x5      x6      x7      x8      x9      x10
x1    0.6125   -0.993    0.1992    0.9421    0.1202    6.938   -8.877    3.475   -10.1   -6.531
x2   -0.05489    0.975   -1.521    0.3005    0.5712    8.546   -11.34    4.186   -11.84   -7.113
x3    -1.119   -0.6407   -0.7236   -0.3731    0.4128   -2.032    2.63   -1.011    2.912    1.842
x4   -0.6264    1.809   -0.5933    0.8155   -0.987    3.46   -4.466    1.724   -4.977   -3.165
x5    0.2495   -1.08    0.4013    0.7989    0.7596   -0.1618    0.4959   -0.01364   -0.1927   -0.514
x6    5.673    1.115    2.956   -1.306   -22    1.878   -10.98    0.7181   -7.851   15.59
x7   -1.338   -2.851    1.185    7.278    5.79    9.83   -7.511    1.481   -18.82   -12.33
x8   -3.889    0.2694   -2.778   -1.888   14.84    0.7384    1.72    1.044    4.427   -12.59
x9   -3.256   -4.008    0.7522    9.819   13.41    6.09    1.351    0.3782   -13.98   -17.56
x10   -2.28   -0.08912   -1.449   -0.4418    8.76    2.368   -0.4948    1.837    1.048   -8.514

B =
      u1      u2      u3
x1    0.1837    2.787   -0.4336
x2    0.2908   -1.167   -0.1685
x3    0.1129   -1.854   -0.2185
x4     0.44   -1.141    0.5413
x5    0.1017   -1.093    0.3893
x6   -2.913   -0.4945   -0.5853
x7   -0.4894    3.713    4.255
x8     2.467   -1.097   -1.243
x9    0.1411    4.964    5.694
x10    1.334   -0.3   -0.3357

C =
      x1      x2      x3      x4      x5      x6      x7      x8      x9      x10
y1    0.7512   -1.283   -1.836    2.227    0.2358   -1.801    3.834   -0.5452    0.3537   -1.833
y2     1.778   -2.329    0.06676   -0.06921    0.2458    3.615   -2.897    2.214   -7.823   -7.388
y3     1.223    0.9019    0.03548   -0.5073    0.07005    3.685   -6.021    1.541   -3.426   -0.4549

D =
      u1  u2  u3
y1    0    0    0
y2    0    0    0
y3    0    0    0
```

Continuous-time state-space model.
Model Properties

```
eig(T_cl.A)
```

```
ans = 10x1 complex
-11.4313 + 0.0000i
-4.5839 + 0.0000i
-0.5577 + 1.5618i
-0.5577 - 1.5618i
-0.4545 + 1.4504i
```

```
-0.4545 - 1.4504i
-2.3205 + 0.0000i
-2.2865 + 0.0000i
-0.6478 + 0.0000i
-1.3498 + 0.0000i
```

```
spectrus21
```

```
spectrus21 = 5×1 complex
-0.5577 + 1.5618i
-0.5577 - 1.5618i
-1.3498 + 0.0000i
-2.2865 + 0.0000i
-11.4313 + 0.0000i
```

```
spectrus12 % respecta principiul separatiei Kalman asta
```

```
spectrus12 = 5×1 complex
-0.4545 + 1.4504i
-0.4545 - 1.4504i
-0.6478 + 0.0000i
-2.3205 + 0.0000i
-4.5839 + 0.0000i
```

```
norm(T_c1,2) % norma 2 a lui TLFi(T,K)
```

```
ans =
56.2734
```

Verificam cu formula ca astea e ok

$$V_x^T V_x = D_{12}^T D_{12}.$$

```
Vx = chol(D12' * D12, 'lower')
```

```
Vx = 2×2
    1.4019    1.7837
         0    1.2080
```

```
norm(D12' * D12 - Vx' * Vx) % paranoia lui Andrei cu chol:)))
```

```
ans =
2.2204e-16
```

$$V_y^T V_y = D_{21}^T D_{21}$$

```
Vy = chol(D21 * D21', 'lower')
```

```
Vy = 2×2
    1.4299   -0.3961
         0    0.2681
```

```
norm(D21 * D21' - Vy' * Vy) % paranoia lui Andrei cu chol:)))
```

```
ans =
4.4409e-16
```

Acum scot $W_x = -V_x F_s$

$$W_x = -V_x * F_s$$

$$W_x = 2 \times 5$$

1.5386	0.4975	1.3462	-5.8956	-7.1365
-5.2462	7.6873	-2.4001	6.1896	2.6880

$$\text{Acum scot } W_y = -V_y K_s^T$$

$$W_y = -V_y * K_s' \text{ \% transpun ca sa aiba sens inmultirea matriceala (e practic } K_s \text{ din EMAR netranspus)}$$

$$W_y = 2 \times 5$$

0.1210	5.5430	-2.3077	7.1807	-0.8213
3.0100	-1.1865	-1.8733	-2.3477	-1.1439

Acum vedem ambele formule pentru problema noastra $\mathcal{H}_2$:

$$\min_{K \text{ stabil}} \|TLFi(T, K)\|_2 = \min_{K \text{ stabil}} \|T_{cl}\|_2 = [tr(B_1^T X_s B_1) + tr(W_x Y_s W_x^T)]^{\frac{1}{2}} = [tr(W_y X_s W_y^T) + (C_1 Y_s C_1^T)]^{\frac{1}{2}}$$

$$\text{norma2_manual_1} = \text{sqrt}(\text{trace}(B1' * X_s * B1) + \text{trace}(W_x * Y_s * W_x'))$$

$$\text{norma2_manual_1} = 56.2734$$

$$\text{norma2_manual_2} = \text{sqrt}(\text{trace}(W_y * X_s * W_y') + \text{trace}(C1 * Y_s * C1'))$$

$$\text{norma2_manual_2} = 56.2734$$